# Spatial downscaling for ground level Ozone concentration extremes
# Final project report

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**Abstract**

In this work a two-step approach is proposed to perform downscale of spatial extremes of ground measurements of Ozone concentrations derived from satellite observations at coarser resolution using latent gaussian models. Inference is performed on seasonality corrected data, and computations of quantities like return levels is made by adding back an estimated seasonal model. We found that this approach is computationally feasible, and provides insights about the extreme behavior of the data across the study region. Our results indicate that the ozone levels might not exhibit a heavy-tail behavior. In addition to that, May of 2018 was identified as a month whose return levels were exceeded.

## 1 Introduction

Current remote sensing methods for data collection allow recording information from large spatial regions. Compared to ground measurements, satellite observations are capable of delivering information over large spatial domains. However, they pose some challenges for statistical modeling as the measurements do not correspond to point measurements but to grid measurements, usually as average over regions. These limitations motivate the use of spatial downscaling models to address the spatial resolution issue of satellite measurement. These models can provide with high resolution estimates of different quantities of interest like return levels. Despite these challenges from a modeling perspective, these methods of data collection have proved to be useful in environmental problems like air pollution monitoring. Ozone is one contaminant that poses significant health challenges when exposed to large quantities over a prolonged period, and therefore its monitoring is important to help decision makers create policies that can reduce its impact on human health. In this project, I propose the use of a two-step modeling approach to perform disaggregation and obtain inference at high resolutions for quantities of interest in extremes. This approach models seasonality of the data in the first step across the whole region of study and *corrects* for such seasonality in the data. Later, a latent gaussian model is used to model the spatial dependence and to perform downscaling.

## 2 Preliminary analysis

### 2.1 Description of the dataset

The dataset contains averaged Ozone (O3) concentration levels at ground level for the UK from 2013 to 2022 over a regular grid, these values are derived from satellite observations. This dataset was obtained from the Copernicus Atmosphere Monitoring Service (CAMS) and corresponds to the Reanalysis dataset published in 2025 CAMS (2025).

### 2.2 Exploratory data analysis

For computational reasons, the data comprises only a subregion of the UK. The corresponding polygon is shown in Figure 1b. This region contains 193 grid cells and consists of ozone level

measurements every 3 hours from 2013-2022, given the small number of observations for blocks of one day, blocks of one month were considered. With monthly blocks, an average of 232 observations were included in each block to compute the maximum, instead of only 8 observations for daily blocks. For our analysis, each grid cell had a time series of monthly maxima for 120 months, see Figure 2.

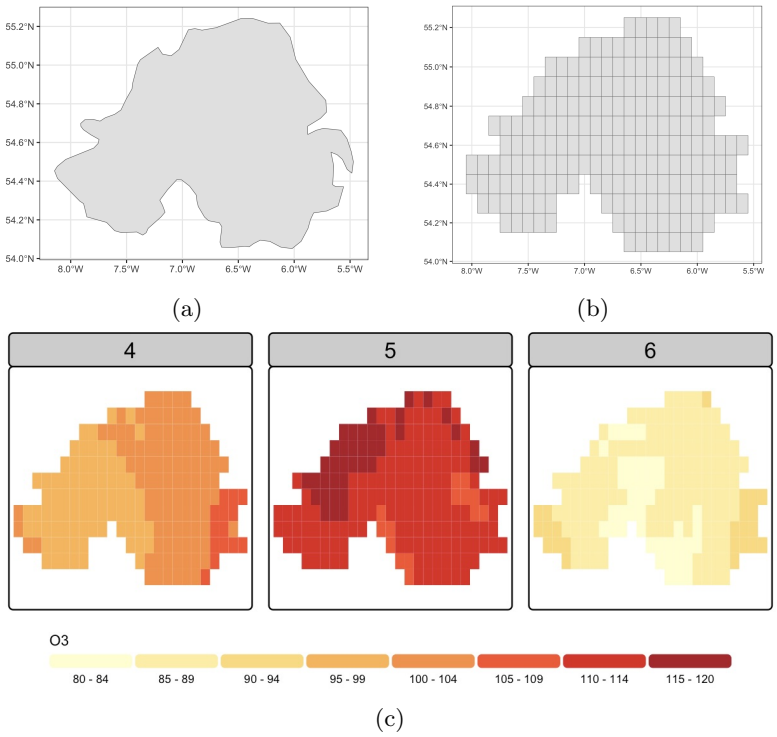



Figure 1: Representation of the region of interest. (a) Polygon of the region of interest, (b) the grid cells at which data was obtained, and c) maxima for April, May and June 2018.

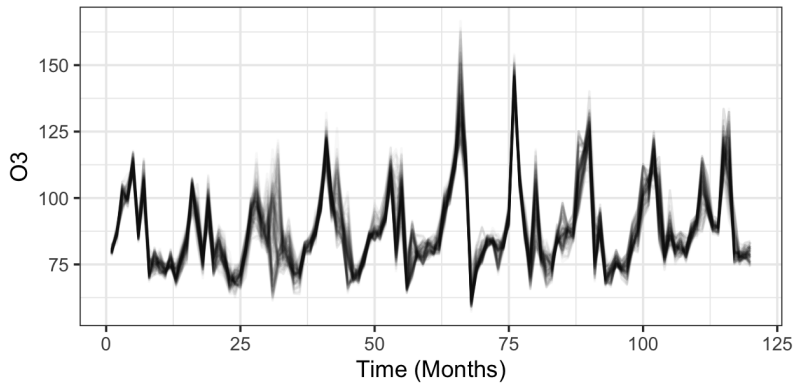
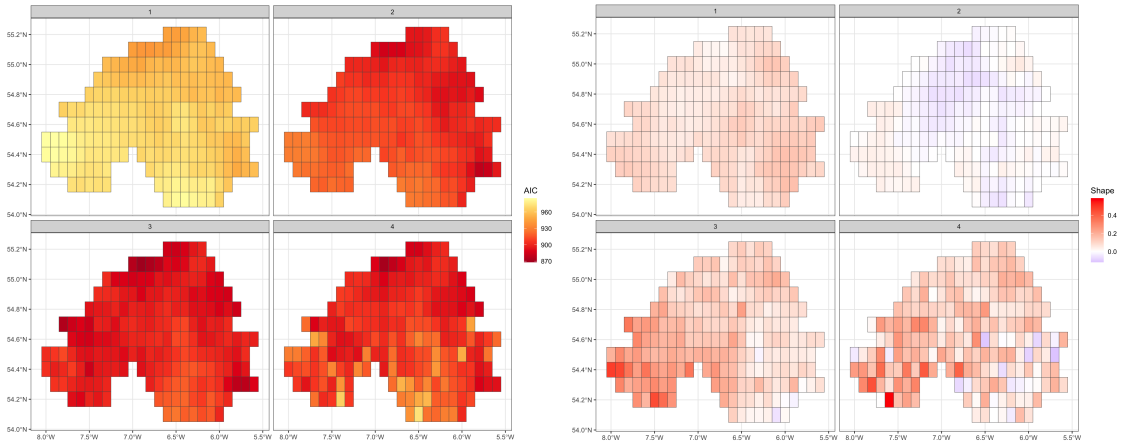



Figure 2: Monthly-maxima time series for the cells inside the region considered.

An important assumption in our proposed approach is that the shape parameter $\xi > 0$. In order to check for the validity of this assumption, individual models for time series were fitted using functions from the `ismev` package. Four different models were tested, the first model included only an intercept, the second model included a pair of sine and cosine terms to account for the seasonal behavior of the data, an intercept term and a slope. The third model, similar to the second, includes an additional set of sine and cosine terms. The fourth model included an extra intercept and slope for the tail parameter, where the slope was associated to the geographical longitude of the grid cell centroid. In order to choose a model, maps of AIC for model were obtained and

displayed in Figure 3a. From the maps we can quickly note that model 2 and model 3 are our best options across the whole region of study. Furthermore, model 3 appears to be the best in most of the region, and therefore this model was chosen for further analysis.




(a) (b)

Figure 3: (a) AIC values for each model on the grid cell data, and (b) the fitted tail ($\hat{\xi}$) parameter.

In addition to that, estimates of the tail parameters were obtained and maps are displayed in Figure 3b, where we can observe that most of the values are positive. Therefore, we expect the assumption of $\xi > 0$ to be reasonable for the data.

# 3 Models and inference

The approach in this project will be Bayesian, and the computations will be handled by the R-INLA package, which provides fast approximate Bayesian inference for latent Gaussian models.

## 3.1 Bayesian inference with R-INLA

In general, the R-INLA package fits models with latent Gaussian effects. The response is assumed to follow a known density and conditional on the latent effects, $\boldsymbol{x}$, the vector of observations are conditionally independent, therefore the likelihood of the model can be written as

$$\boldsymbol{y}|\boldsymbol{x}, \theta = \Pi_{i\in\mathcal{I}}\pi(y_i|\eta_i, \theta) \tag{1}$$

where $\eta_i$ is linear predictor obtained as linear combinations of latent effects. In the INLA methodology, it is of central interest the computation of posterior marginals of the model effects and the hyperparameters. Computationally, this is obtained by considering the computational properties of Gaussian Markov Random Fields (GMRFs), and the Laplace approximation for integration in multiple dimensions. In this case, the joint posterior of the latent effects and their corresponding hyperparameters can be expressed as follows

$$\begin{aligned}\pi(\boldsymbol{x}, \theta) &\propto \pi(\theta)\pi(\boldsymbol{x}|\theta)\Pi_{i\in\mathcal{I}}\pi(y_i|\eta_i, \theta) \\ &\propto \pi(\theta)|Q(\theta)|^{1/2}\exp\left\{-\frac{1}{2}\boldsymbol{x}^T Q(\theta)\boldsymbol{x} + \sum_{i\in\mathcal{I}}\log\pi(y_i|\eta_i, \theta)\right\},\end{aligned} \tag{2}$$

where $Q(\theta)$ represents the precision matrix of the latent effects $\boldsymbol{x}$ parameterized by $\theta$, and $|Q(\theta)|$ its the determinant of $Q(\theta)$. Then, computation of marginal posteriors can be done for the latent effects and the hyperparameters by noting that

$$\pi(x_i|\boldsymbol{y}) = \int \pi(x_i|\theta, \boldsymbol{y})\pi(\theta)d\theta \tag{3}$$

$$\pi(\theta_j|\boldsymbol{y}) = \int \pi(\theta|\boldsymbol{y})d\theta_{-j} \tag{4}$$

where $d\theta_{-j}$ stands for the *differential* of the vector of hyperparameters $\theta$, except the $j-th$ component. Several approximations are employed in order to compute the two integrals above, further details can be found in the original paper of the INLA methodology Rue et al. (2009).

## 3.2 The bGEV distribution

The blended Generalized Extreme Value distribution Castro-Camilo et al. (2022), bGEV, is an alternative distribution to the generalized extreme value distribution when the tail parameter is positive, i.e., $\xi > 0$. This distribution is defined as a combination of the Frechet and Gumbel distributions and it is defined below

$$H(x|\theta) = F(x|q_{\alpha_F}, s_{\beta_F}, \xi)^{p(x|s_{p_1}, s_{p_2}, a, b)} G(x|q_{\alpha_G}, s_{\beta_G})^{1-p(x|s_{p_1}, s_{p_2}, a, b)}. \tag{5}$$

In the equation above, $F(\cdot)$ and $G(\cdot)$ are the Frechet and Gumbel CDFs, respectively; while $p(x|s_{p_1}, s_{p_2}, a, b)$ is the CDF of a Beta distribution with shape parameters $s_{p_1} > 1$, $s_{p_2} > 1$, evaluated at the point $(x-a)/(b-a)$, and it controls how the distributions F and G influence the model H.

## 3.3 Two step approach for downscaling of spatial extremes

A reasonable assumption for this dataset is that the observed gridded satellite data are an aggregated realization of a continuous latent random field. Then, we can employ the SPDE approach implemented in R-INLA package to fit the models. In INLA, the GEV distribution is not available due to the dependence of the distribution's support on its parameters, which causes numerical issues. Therefore, the bGEV distribution was used as likelihood for the response.

Given the complexity of the combination of the downscaling task and the chosen likelihood for the data, a two step approach for modeling the data was implemented. At the first step, a model to account for seasonality across all grids was fitted, this step is justified by the fact that seasonality was observed in the time series plots of the data in each cell. Then, the data was corrected using the fitted model for seasonality. For the second step a model with a spatial component for the location was fitted. In the next two section, details are provided for each step

### 3.3.1 Seasonality correction

In the following, $y_{mtk}$ will denote the monthly maxima for the $m$-th month of year $t$ in the grid $k$. In our data $m = 1, \ldots, 12$, $t = 1, \ldots, 10$, and $k = 1, \ldots, 193$.

The first step in our modeling approach is a correction for the seasonality observed in the times series for each grid cell. The correction was obtained by first fitting a non-stationary model to the time series within the region of interest. This approach facilitates adding back the seasonal component for computation of return levels, if cell-level models were fitted, then we will be able to add the seasonal component back only to those regions encompassed by the available cells and we will potentially do not cover the complete region. The model is specified in Equation 6.

$$\begin{aligned} y_{mtk} &\overset{iid}{\sim} GEV(\mu_m, \sigma, \xi), \quad i = 1, \ldots, 10, j = 1, \ldots, 12, k = 1, \ldots, 193. \\ \mu_m &= \beta_0 \cdot t + \sum_{i=1}^{2} \left( \sin\left(\frac{2\pi i(m-1)}{12}\right) + \cos\left(\frac{2\pi i(m-1)}{12}\right)\right) \end{aligned} \tag{6}$$

The resulting fit is displayed in Figure 5a. We can observe that the seasonal component captures quite reasonable the seasonal behavior observed in the time series, while it does not over-smooth the data; this is particularly important for the analysis of extremes since over-smoothing can potentially mask the extreme behavior.

The seasonality-corrected data, denoted by $\tilde{y}_{mtk}$, is defined by $\tilde{y}_{mtk} = y_{mtk} - \hat{\mu}_m$, where $\hat{\mu}_m$ is the fitted location obtained by fitting the model in Equation 6.

### 3.3.2 Spatial model for corrected data

Let $B_k$, $k = 1, \dots 193$, denote the region spanned by the $k-$th cell in the grid, the model for spatial modeling of the seasonality-corrected data is described by Equation 7.

$$\begin{aligned} \tilde{y}_{mtk}|\boldsymbol{\beta}(B_k) &\sim \text{bGEV}(\mu_{mtk}, \sigma, \xi) \\ \log(\tilde{\mu}_{mtk}) &= \frac{1}{|B_k|}\int_{B_k} \boldsymbol{\beta}(\mathbf{s}) d\mathbf{s} \end{aligned} \tag{7}$$

In the model structure above, we assume that after correcting for seasonality, the temporal dependence is mild so it can be ignored and the focus is on spatial dependence in the location. The location of the seasonality corrected data is modeled as the spatial average of a spatially continuous latent gaussian component over the corresponding grid cell. This latent gaussian model $\boldsymbol{\beta}(\boldsymbol{s})$ continuous in space, is the solution to the following stochastic partial differential equation

$$\left(\kappa^2 - \Delta\right)^{\alpha/2} \boldsymbol{\beta}(\boldsymbol{s}) = \mathcal{W}(\boldsymbol{s}), \tag{8}$$

where $\boldsymbol{\beta}(\boldsymbol{s})$ is the spatial field, $\kappa =$ inverse range parameter, $\alpha = \nu + d/2$ (with spatial smoothness $\nu$, dimension $d$), $\mathcal{W}(\boldsymbol{s})$ is spatial white noise, $\Delta$ is the Laplacian. The resulting solution has Matérn covariance

$$\text{Cov}\left(\boldsymbol{\beta}(\boldsymbol{s}), \boldsymbol{\beta}\left(\boldsymbol{s}'\right)\right) = \sigma^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\kappa \left\|s - s'\right\|\right)^{\nu} K_{\nu}\left(\kappa \left\|s - s'\right\|\right)$$

With this spatially continuous process, it is possible to perform inference at any desired resolution after fitting the model. As noted by Moraga et al. (2017), the stochastic integrals appearing in Equation 7 can be approximated by

$$\int_{B_j} \boldsymbol{\beta}(\mathbf{s}) d\mathbf{s} \approx \sum_{g=1}^{G} A_{jg}\beta_g,$$

in the SPDE approach implemented in R-INLA; where $G$ is the number of vertices in the triangulated mesh for the region of interest, and $A$ is a $J \times G$ sparse matrix that links the components of the GMRF to the $J$ gridded observations. For our gridded data, the vertices of the triangulated mesh are displayed in Figure 4. On average, 5 nodes lie inside of each grid cell. The number of vertices will influence the computational load implied by the model, the higher the number of vertices per grid cell, the more computationally expensive. Given that this model was fitted in a personal computer, this number of vertices allows obtaining results in a reasonable amount of time.

As with any bayesian model, prior distribution must be set for the hyperparameters of the model. For all the components of this model, penalized complexity priors were chosen, and are given by default priors in R-INLA. It is important to remark that the penalized complexity prior for the tail parameter of the bGEV distribution corresponds to the Gumbel model, i.e., $\xi = 0$. Therefore, it is expected that if the data contains enough evidence for deviations from this model, it will be expressed in the shape of the posterior density for such parameter.

## 3.4 Posterior inference

After obtaining the posterior distributions from the model of seasonality-corrected data, we can obtain posterior inference for different quantities of interest. Of particular interest are posterior return levels, these can be obtained as follows:

$$\pi(y_{mtk,R}(\boldsymbol{s})) = \int_{\theta} \pi(y_{mtk,R}(\boldsymbol{s})|\mu_t, \sigma, \xi)\pi(\mu_t, \sigma, \xi|\mathbf{y}, \boldsymbol{\beta}(\boldsymbol{s}))d\theta, \tag{9}$$

where $y_{mtk,R}$ is the $R$-return level. Given that the posterior distribution is computed for seasonality-corrected data, we must add-back to the location the seasonality term denoted here by $s_m$, then

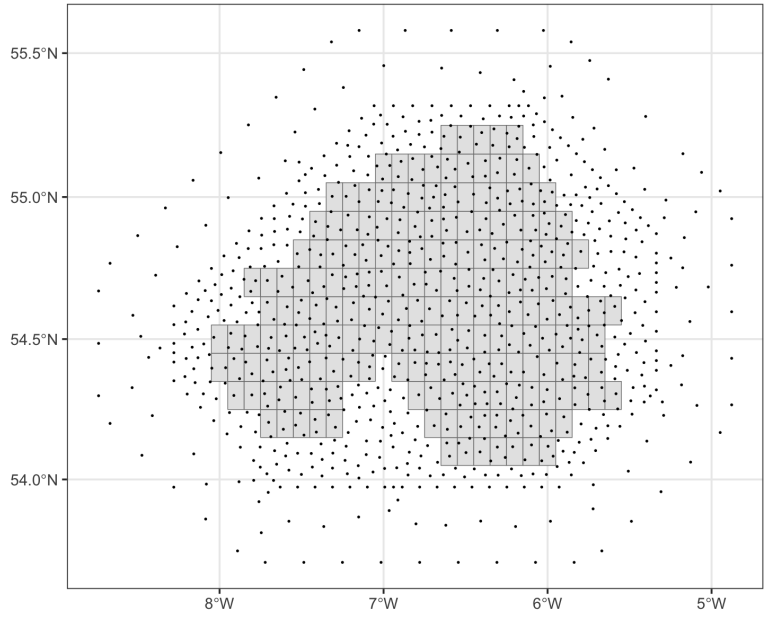



Figure 4: Mesh nodes for the definition of the spatial SPDE model. The points inside each cell will be used as integration point in the definition of the $\log(\tilde{\mu}_{mtk})$

the location is set to $\mu_{mkt} = s_t + \tilde{\mu}_{mtk}$, with $\tilde{\mu}$ as the posterior location parameter of the bGEV model to the seasonality-corrected data. This distribution can be obtained by drawing samples from the estimated posterior distribution from the data, and then producing summaries to study its behavior.

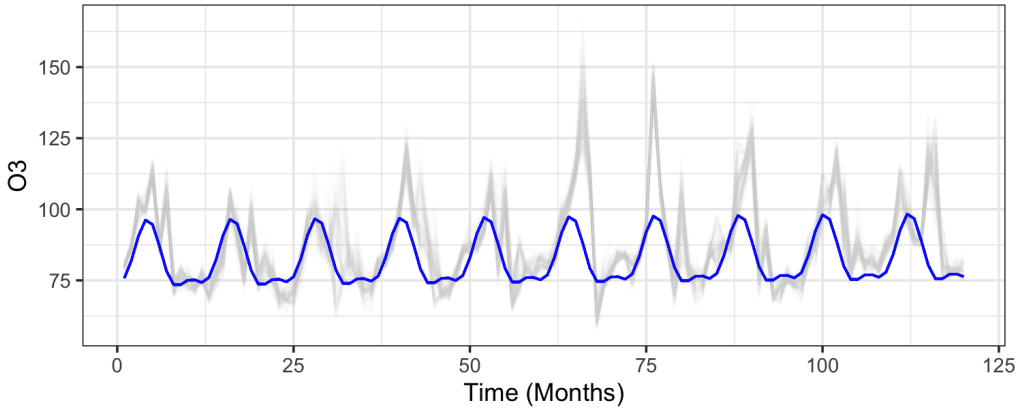
# 4 Results

## 4.1 Seasonality correction

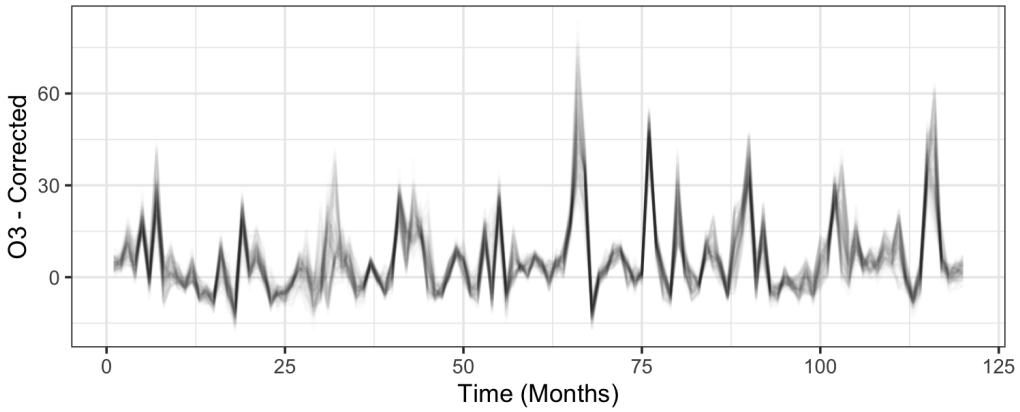
Our first results correspond to the fitted seasonal component. The estimated seasonal location is displayed in Figure 5a as a blue line, and the original time series for each grid cell in light gray. We can observe that the fitted seasonal component captures well the overall behavior of the time series. It does not overfit the observations, which is a good sign for further modeling of extremes since we expect the tail behavior to not being removed from the seasonality corrected data. We can also observe that there is no apparent residual trend in the corrected data, so there was no need to include a component for that in the spatial model. An unexpected behavior was found at the time of inspecting autocorrelation plots, it was found that autocorrelation was mild even at lag-1.

## 4.2 Posterior distribution results

The shape of the posterior marginal distribution for the model hyperparameters are displayed in Figure 6. In the first row (panel a) we find the posterior marginals of the likelihood component, i.e., $\sigma$ and $\xi$. We can observe that the marginal posterior for spread $\sigma$ is quite symmetric around 2.10. This means that the observations at the data level are moderately variable. Regarding the tail parameter $\xi$ that describes the tail behavior, we can observe that the marginal posterior is very concentrated on very small values. This points to the tail parameter being 0 very likely and




(a) Seasonality component




(b) Seasonality corrected data

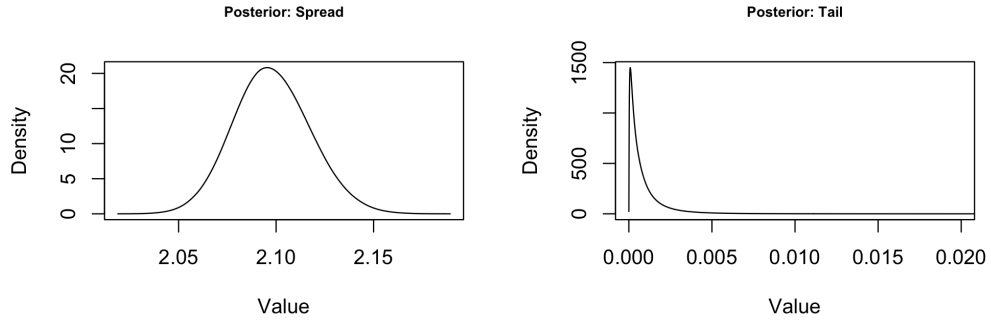
Figure 5: a) Fitted seasonality component (blue) for the time series data (original data in light gray color), and b) seasonality corrected Ozone data at each grid cell after removing the fitted seasonal location.

therefore indicate that the data does not exhibits heavy tailed behavior. As the prior for such parameter is related to the Gumbel model, we can observe that the data does not provide enough evidence to deviate from such base model.

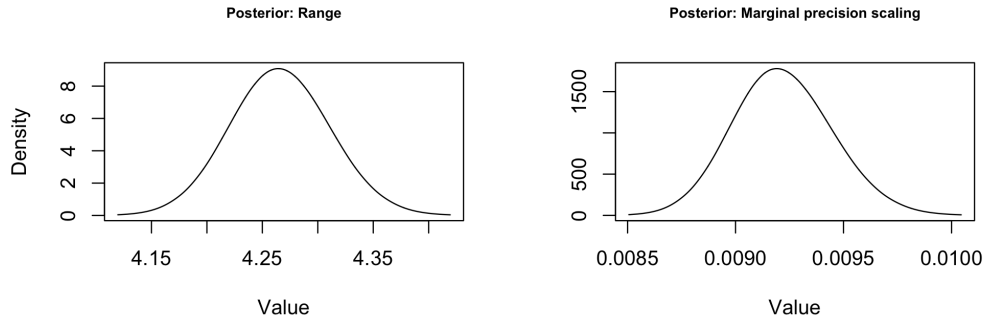
For the spatial component, we have that the marginal distribution of the posterior range $\kappa$ is centered around 4.3. This implies that the spatial range is not so big for our data and that more localized behavior is expected for the latent process. On the other hand, the marginal posterior for the precision scaling $\tau$ concentrates its mass around small values, which indicates that the marginal variance of the process is high. Table 1 summarizes quantiles from each marginal posterior distribution.

| Parameter | $q_{0.025}$ | $q_{0.5}$ | $q_{0.975}$ |
|---|---|---|---|
| $\sigma$ | 2.0604 | 2.0974 | 2.1386 |
| $\xi$ | 0.0000 | 0.0005 | 0.0044 |
| $\kappa$ | 4.1783 | 4.2655 | 4.3559 |
| $\tau$ | 0.0088 | 0.0092 | 0.0097 |

Table 1: Posterior marginal summaries for hyperparameters of the model components




(a) Posterior distribution for bGEV likelihood.




(b) Posterior distribution for spatial SPDE.

Figure 6: Posterior distributions of model hyperparameters of the seasonality-corrected data.

### 4.3 Posterior return levels

For our data, the 10-years return levels were computed for the months of April, May and June of 2018. These months were chosen based on Figure 5a. In that Figure, the chosen months appear to have recorded more high ozone concentrations compared to previous years. The maps in Figure 7 show the quantiles of the posterior distribution of the return level for each month.

We can observe that in April, the return level appears to be higher than those of the following two months. An interesting behavior is observed in that *transition*. For April, the map show a quite uniform spatial patter across the region; later in May the map shows that around half of the region has started to decrease in the maxima of ozone levels, and finally in June the pattern appears to have decreased across all the region.

When we observe the patterns at the lower tails of the posterior distribution for the return level of each month, we can note that for April and June there are higher values of ozone than in the *borders*. For may, the pattern appears to be quite similar across all the region. For the upper tail, we can observe a similar *transition* pattern as for the median; with April being the month with more high return levels, May being the *transition* month, and June being the month at which the pattern of return levels appears to be similar across the region.

Comparing the maps in Figure 1c with those of the computed summaries for the return levels, we can immediatly see that May, 2018 can be flagged as month with Ozone levels more extremes than those expected by the model. For the month of April, 2018, some regions could also be flagged but not the entire region.

## 5 Conclusions

In this project I explored the use of latent Gaussian models to perform spatial downscaling for extreme Ozone level using 10 years of monthly maxima using a two step approach to tackle nu-

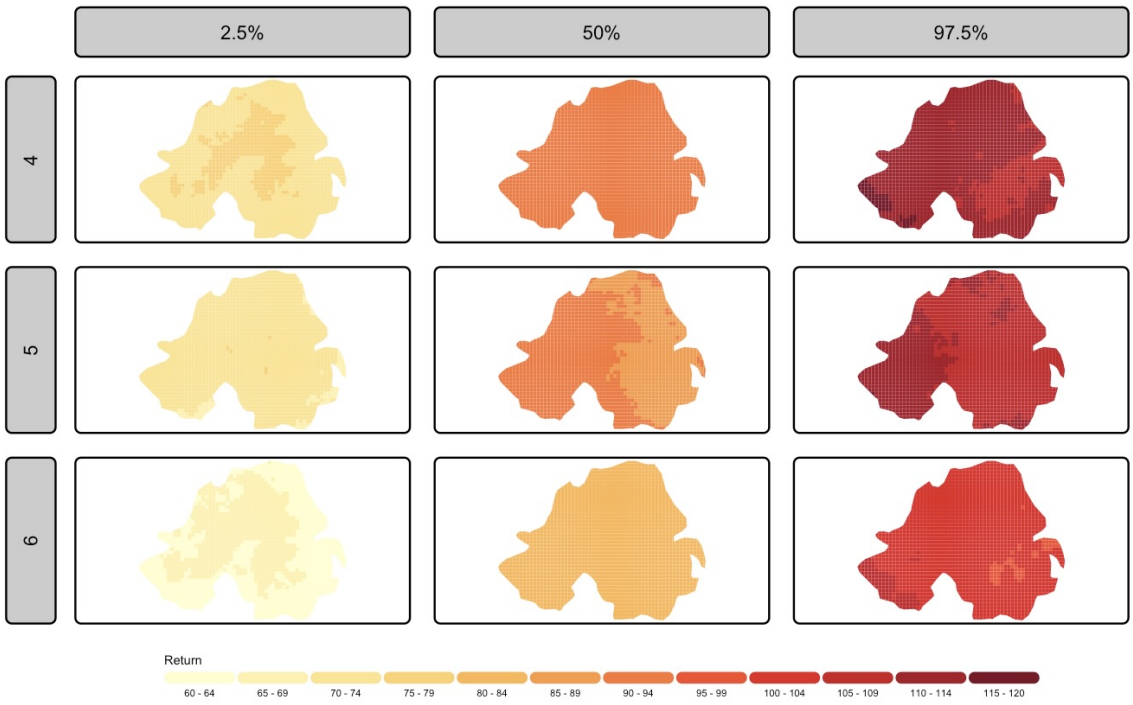



Figure 7: Posterior 10-years return levels with 2.5%, 50%, 97.5% quantiles for April, May and June of 2018

merical issues. The results indicate that the latent spatial process exhibits localized variations and high variability. For the bGEV likelihood, the results indicate small variability, and evidence that Ozone concentration in this regions may not exhibit heavy-tails. First, the two-step approach faces several limitations, specially in terms of integration of uncertainty and *efficient* use of the data. Our correction for seasonality is done using likelihood methods and independent of the spatial modeling; therefore we could ask whether we are making the best use of the data for estimation; in addition, the uncertainty of the estimated seasonal component is not integrated in the computation of return levels and the posterior distribution can potentially underestimate uncertainty. Secondly, we could also argue that a joint model could be a better option to integrate all sources of uncertainty. However, when tried to fit a joint model, I encountered several issues. The main source of those issues was at the likelihood level, i.e., the bGEV density. As explained by Castro-Camilo et al. (2022), the bGEV distribution lacks of the so called *log-concavity*, and several approaches were developed by the authors to avoid the restriction in the domain, which otherwise causes more issues to INLA. Further developments can be made for this work, in principle, joint modeling will be ideal. Possible solutions for the numerical issues include setting more informative priors for the scale and tail parameters. In addition of the join modeling, covariates can be included to account for Ozone drivers like temperature or wind direction.

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